

ASTM SPECIFICATION & TOLERANCE FOR TUBING & PIPING

Specification	Allowable Outside Diameter Variation in mm			Allowable Wall Thickness Variation		Exact Length Tolerance in mm		Testing
ASTM-A213 Seamless Boiler Superheater and Heat Exchanger Tubes	Nominal Diameter	Over	Under	%Over	%Under	Over	Under	Flattening Test Tension Test Flare Test Hardness Test 100% Hydrostatic test Refer to ASTM A-450
	Under 25.4	.1016	.1016	+20	-0	3.175	0	
	25.4-38.1 incl	.1524	.1524	+22	-0	3.175	0	
	38.1-50.8 excl	.2032	.2032	+22	-0	3.176	0	
	50.8-63.5 excl	.254	.254	+22	-0	4.46	0	
	63.5-76.2 excl	.3048	.3048	+22	-0	4.76	0	
ASTM-A249 Welded Boiler Superheater, Heat Exchanger And Condenser Tubes	76.2-101.6 incl	.381	.381	+22	-0	4.76	0	Tension Test Flattening Test Flange Test Reverse Bend Test Hardness Test 100% Hydrostatic Test Refer to ASTM A-450
	Under 25.4	.1016	.1016	+10	-10	3.175	0	
	25.4-38.1 incl	.1524	.1524	+10	-10	3.175	0	
	38.1-50.8 Excl	.2032	.2032	+10	-10	3.175	0	
	50.0-63.5 excl	.254	.254	+10	-10	4.762	0	
	63.5-76.2 excl	.3848	.3048	+10	-10	4.762	0	
ASTM-A269 Seamless & Welded Tubing for General Service	76.2-101.6 incl	.381	.381	+10	-10	4.762	0	Flare Test (Seamless Only) Flange Test (Welded Only) Hardness Test Reverse Flattening Test (Welded only) 100% Hydrostatic Test Refer to ASTM-A269
	Untp 12.7	.13	.13	+15	-15	3.2	0	
	12.7-38.1 excl	.13	.13	+10	-10	3.2	0	
	38.1-88.9 excl	.25	.25	+10	-10	4.8	0	
	88.9-139.7 excl	.38	.38	+10	-10	4.8	0	
ASTM-A270 Seamless & Welded Sanitary Tubing	139.7-203.2 excl	.76	.76	+10	-10	4.8	0	Reverse flattening Test 100% Hydrostatic Test External' polish on all tubes Refer to ASTM A-270
	25.4	.05	.20	+10.0	-10.0	3.2	0	
	38.1	.05	.20	+10.0	-10.0	3.2	0	
	50.8	.05	.28	+10.0	-10.0	3.2	0	
	60.5	.05	.28	+10.0	-10.0	3.2	0	
	76.2	.08	.30	+10.0	-10.0	3.2	0	
ASTM-A312 Seamless & Welded pipe	101.6	.08	.38	+10.0	-10.0	4.8	0	Tension Test Flattening Test 100% Hydrostatic Test Refer to ASTM A-530
	3.175-38.1 incl	.4	.79	Minimum Wall 12.5% under nominal wall Specified		6.4	0	
	38.1-101.6 incl	.79	.79			6.4	0	
ASTM A-358 Welded pipe	101.6-203.2 incl	1.59	.79	-0.3		6.4	0	Refer to ASTM A-530
	219.08-750mm	+0.5%				6.0		